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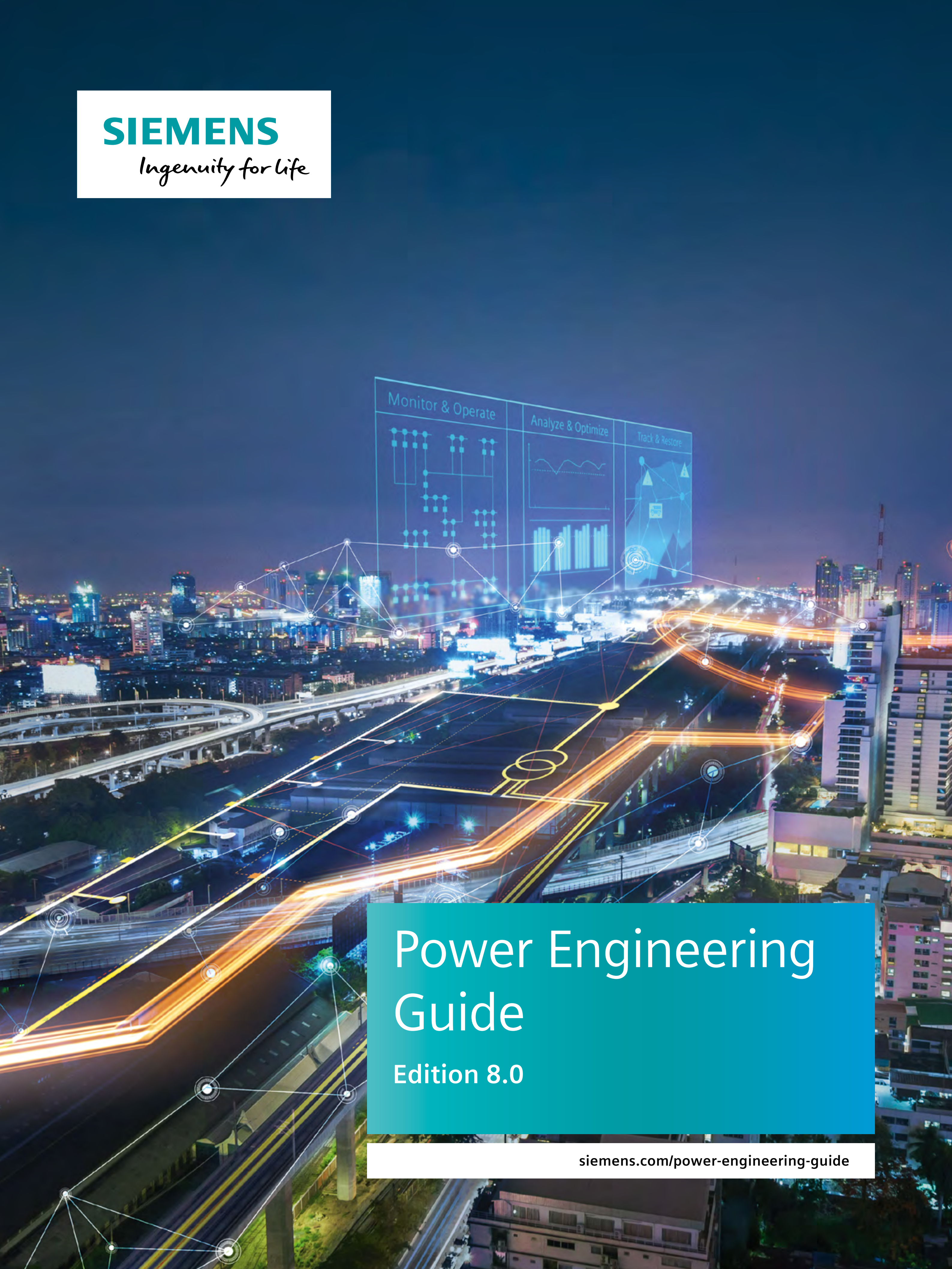

Power Engineering Guide

Edition 8.0

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Power Engineering Guide

Edition 8.0

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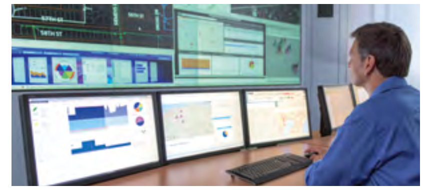

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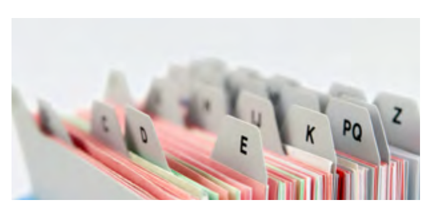

Abbreviations, trademarks

5.11 Distribution transformers

5.11.1 Liquid-immersed distribution transformers for European / U.S. / Canadian standard

On the last transformation step from the power plant to the consumer, distribution transformers (DT) provide the necessary power for systems and buildings. Accordingly, their operation must be reliable, efficient and, at the same time, silent.

Distribution transformers are used to convert electrical energy of higher voltage, usually up to 36 kV, to a lower voltage, usually 250 up to 435 V, with an identical frequency before and after the transformation. Application of the product is mainly within suburban areas, public supply authorities and industrial customers.

Distribution transformers are fault-tolerant, economic, and have a long life expectancy. These fluid-immersed transformers can be 1-phase or 3-phase. During operation, the windings can be exposed to high electrical stress by external overloads, and high mechanical stress by short circuits. They are made of copper or aluminum. Low-voltage windings are made of strip or flat wire, and the high-voltage windings are manufactured from round wire or flat wire.

Three product classes are available, as follows:

- Standard distribution transformers:
  - 1-phase or 3-phase, pole-mounted (fig. 5.11-1) or pad-mounted (fig. 5.11-2), wound or stacked core technology distribution transformer (≤ 2,500 kVA, U_m ≤ 36 kV
  - Medium distribution transformer (> 2,500 ≤ 6,300 kVA, Um ≤ 36 kV)
  - Large distribution transformer (> 6.3 – 30.0 MVA, U_m ≤ 72.5 kV)
- Special distribution transformers:
  - Special application: self-protected DT, regulating DT, low-emission DT, or others (autotransformer, transformer for converters, double-tier, multi-winding transformer, earthing transformer)
  - Environmental focus: amorphous core DT with significantly low no-load losses, DT with especially low load-loss design, low-emission DT in regard of noise and/or electromagnetic field emissions, DT with natural or synthetic ester where higher fire resistance and/or biodegradability is required
- Renewable distribution transformers:
  - Used in wind power plants, solar power plants, or sea flow/generator power plants.

Fig. 5.11-1: 1-phase pole-mounted LiDT

Fig. 5.11-2: 3-phase DT, pad-mounted

Liquid-immersed distribution transformer selection table – Technical data, dimensions and weights

EcoDesign and A0Bk loss levels

Rated power * EcoDesign	Primary		Tapping	Secondary		Vector group	Max. no-load losses	Max. load losses 75 °C	Impedance voltage	Sound power level	Sound pressure level	Dimensions				
	No-load voltage	Insulation level		No-load voltage	Insulation level							Length	Width	Height	Distance between rollers	Weight
100 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	145 W	1,750 W	4%	41 dB(A)	29 dB(A)	900 mm	650 mm	1,300 mm	520 mm	500 kg
100 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	145 W	1,475 W	4%	41 dB(A)	29 dB(A)	1,000 mm	750 mm	1,350 mm	520 mm	700 kg
100 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	145 W	1,750 W	4%	41 dB(A)	29 dB(A)	1,000 mm	750 mm	1,400 mm	520 mm	650 kg
100 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	145 W	1,475 W	4%	41 dB(A)	29 dB(A)	950 mm	700 mm	1,300 mm	520 mm	700 kg
160 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	210 W	2,350 W	4%	44 dB(A)	32 dB(A)	950 mm	850 mm	1,350 mm	520 mm	650 kg
160 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	210 W	2,000 W	4%	44 dB(A)	32 dB(A)	1,000 mm	750 mm	1,650 mm	520 mm	900 kg
160 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	210 W	2,350 W	4%	44 dB(A)	32 dB(A)	1,000 mm	750 mm	1,400 mm	520 mm	750 kg
160 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	210 W	2,000 W	4%	44 dB(A)	32 dB(A)	1,000 mm	750 mm	1,450 mm	520 mm	900 kg
250 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	300 W	3,250 W	4%	47 dB(A)	34 dB(A)	1,150 mm	850 mm	1,500 mm	520 mm	950 kg
250 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	300 W	2,750 W	4%	47 dB(A)	34 dB(A)	1,100 mm	800 mm	1,350 mm	520 mm	1,150 kg
250 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	300 W	3,250 W	4%	47 dB(A)	34 dB(A)	1,150 mm	750 mm	1,650 mm	520 mm	1,150 kg
250 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	300 W	2,750 W	4%	47 dB(A)	34 dB(A)	1,100 mm	800 mm	1,500 mm	520 mm	1,150 kg
400 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	430 W	4,600 W	4%	50 dB(A)	37 dB(A)	1,250 mm	950 mm	1,500 mm	670 mm	1,300 kg
400 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	430 W	3,850 W	4%	50 dB(A)	37 dB(A)	1,250 mm	850 mm	1,400 mm	670 mm	1,500 kg
400 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	430 W	4,600 W	4%	50 dB(A)	37 dB(A)	1,350 mm	1,000 mm	1,700 mm	670 mm	1,500 kg
400 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	430 W	3,850 W	4%	50 dB(A)	37 dB(A)	1,250 mm	850 mm	1,450 mm	670 mm	1,500 kg
630 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	600 W	6,500 W	4%	52 dB(A)	38 dB(A)	1,350 mm	1,100 mm	1,650 mm	670 mm	1,850 kg
630 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	600 W	5,400 W	4%	52 dB(A)	38 dB(A)	1,350 mm	1,050 mm	1,700 mm	670 mm	2,200 kg
630 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	600 W	6,500 W	4%	52 dB(A)	38 dB(A)	1,350 mm	1,000 mm	1,850 mm	670 mm	2,150 kg
630 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	600 W	5,400 W	4%	52 dB(A)	38 dB(A)	1,250 mm	850 mm	1,650 mm	670 mm	2,200 kg
630 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	600 W	6,500 W	6%	52 dB(A)	38 dB(A)	1,450 mm	900 mm	1,650 mm	670 mm	1,850 kg
630 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	560 W	5,600 W	6%	52 dB(A)	38 dB(A)	1,450 mm	950 mm	1,700 mm	670 mm	2,100 kg
630 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	600 W	6,500 W	6%	52 dB(A)	38 dB(A)	1,450 mm	1,000 mm	1,800 mm	670 mm	2,150 kg
630 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	560 W	5,600 W	6%	52 dB(A)	38 dB(A)	1,300 mm	850 mm	1,650 mm	670 mm	2,250 kg
800 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	650 W	8,400 W	6%	53 dB(A)	39 dB(A)	1,850 mm	1,150 mm	1,700 mm	670 mm	2,250 kg
800 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	650 W	7,000 W	6%	53 dB(A)	39 dB(A)	1,600 mm	1,000 mm	1,650 mm	670 mm	2,650 kg
800 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	650 W	8,400 W	6%	53 dB(A)	39 dB(A)	1,700 mm	1,050 mm	1,900 mm	670 mm	2,750 kg
800 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	650 W	7,000 W	6%	53 dB(A)	39 dB(A)	1,600 mm	1,000 mm	1,650 mm	670 mm	2,650 kg
1,000 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	770 W	10,500 W	6%	55 dB(A)	41 dB(A)	1,950 mm	1,150 mm	1,850 mm	820 mm	2,850 kg
1,000 kVA	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	770 W	9,000 W	6%	55 dB(A)	41 dB(A)	1,900 mm	1,050 mm	1,950 mm	820 mm	3,300 kg
1,000 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	770 W	10,500 W	6%	55 dB(A)	41 dB(A)	1,700 mm	1,200 mm	2,000 mm	820 mm	3,100 kg
1,000 kVA	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	770 W	9,000 W	6%	55 dB(A)	41 dB(A)	1,900 mm	1,050 mm	1,950 mm	820 mm	3,300 kg
1,250 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	950 W	11,000 W	6%	56 dB(A)	42 dB(A)	2,000 mm	1,100 mm	2,100 mm	820 mm	3,900 kg
1,250 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	950 W	11,000 W	6%	56 dB(A)	42 dB(A)	1,950 mm	1,200 mm	2,050 mm	820 mm	3,750 kg
1,600 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	1,200 W	14,000 W	6%	58 dB(A)	43 dB(A)	2,100 mm	1,300 mm	2,100 mm	820 mm	4,600 kg
1,600 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	1,200 W	14,000 W	6%	58 dB(A)	43 dB(A)	1,950 mm	1,250 mm	2,100 mm	820 mm	4,450 kg
2,000 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	1,450 W	18,000 W	6%	60 dB(A)	45 dB(A)	2,200 mm	1,300 mm	2,250 mm	1,070 mm	5,700 kg
2,000 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	1,450 W	18,000 W	6%	60 dB(A)	45 dB(A)	2,200 mm	1,300 mm	2,250 mm	1,070 mm	5,550 kg
2,500 kVA*	10 kV	LI 75 AC28	± 2 × 2.5%	400 V	LI - AC3	Dyn5	1,750 W	22,000 W	6%	63 dB(A)	48 dB(A)	2,200 mm	1,300 mm	2,200 mm	1,070 mm	5,750 kg
2,500 kVA*	20 kV	LI 125 AC50	± 2 × 2.5%	400 V	LI - AC3	Dyn5	1,750 W	22,000 W	6%	63 dB(A)	48 dB(A)	2,150 mm	1,300 mm	2,200 mm	1,070 mm	5,800 kg

Table 5.11-1: Liquid-immersed distribution transformer selection table – Technical data, dimension and weights

1 2 3 4 5 6 7 8 9 10 11 12 ?

5.11.3 GEAFOL cast-resin transformers

GEAFOL transformers have been in successful service since 1965. Many licenses have been granted to major manufacturers throughout the world since then. Over 100,000 units have proven themselves in power distribution or converter operation all around the globe.

Advantages and applications

GEAFOL distribution and power transformers in ratings from 100 to approximately 50,000 kVA, and lightning impulse (LI) values up to 250 kV are full substitutes for liquid-immersed transformers with comparable electrical and mechanical data. They are designed for indoor installation, close to their point of use at the center of the major load consumers. The exclusive use of flame-retardant insulating materials frees these transformers from all restrictions that apply to oil-filled electrical equipment, such as the need for oil collecting pits, fire walls, fire extinguishing equipment. For outdoor use, specially designed sheet-metal enclosures are available.

GEAFOL transformers are installed wherever oil-filled units cannot be used, or where use of liquid-immersed transformers requires major constructive efforts, for example, inside buildings, in tunnels, on ships, cranes and offshore platforms, inside wind turbines, in groundwater catchment areas, in food processing plants, and in portable or static containers.

Often, these transformers are combined with their primary and secondary switchgear and distribution boards in compact substations that are installed directly at their point of use.

When used as static converter transformers for variable speed drives, they can be installed together with the converters at the drive location. This reduces construction requirements, cable costs, transmission losses, and installation costs.

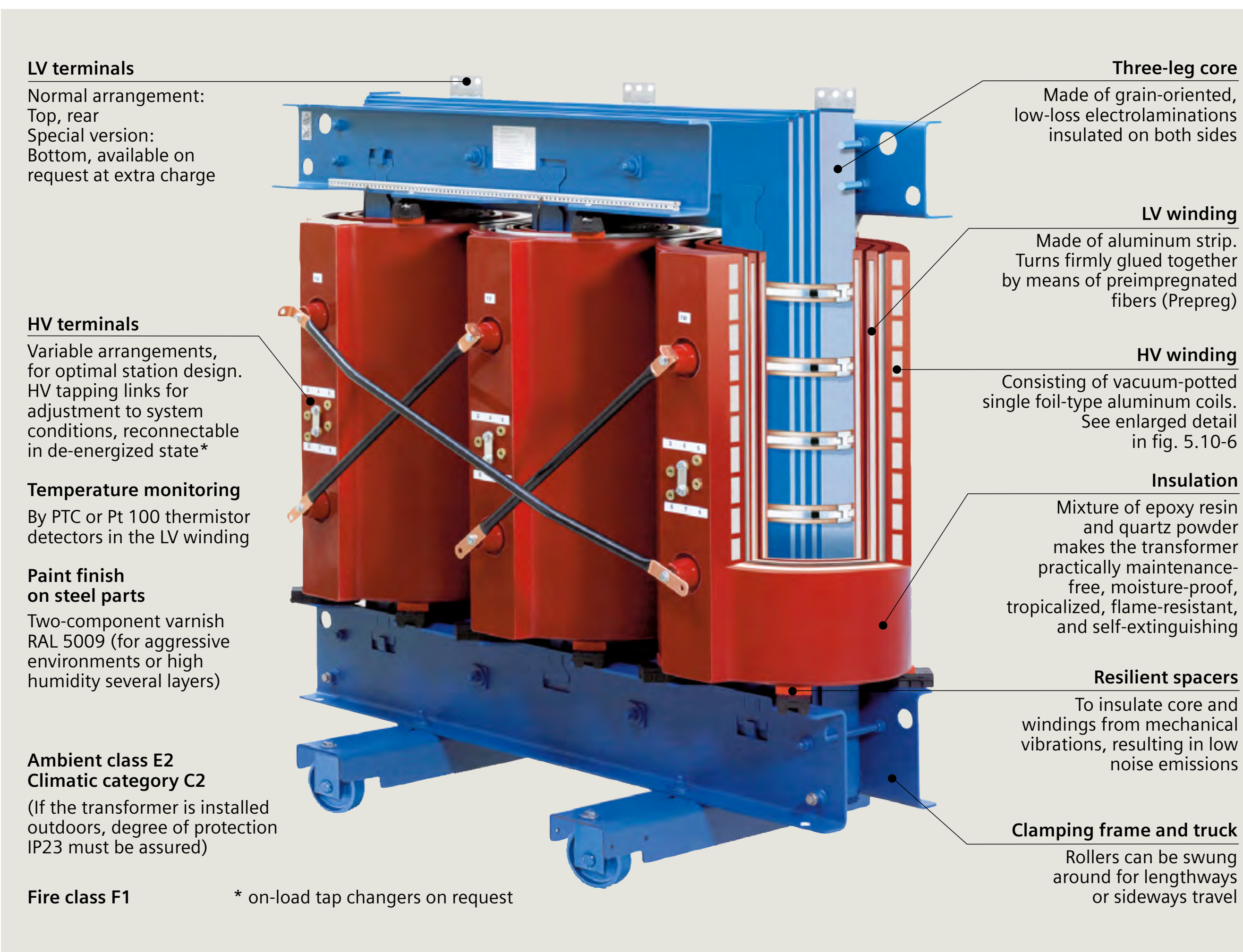

Fig. 5.11-5: GEAFOL cast-resin dry-type transformer properties

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GEAFOL transformers are fully BIL-rated. Their noise levels are comparable to oil-filled transformers. Taking into account the indirect cost reductions, they are cost-competitive. By virtue of their design, GEAFOL transformers are practically maintenance-free.

Standards and regulations

GEAFOL cast-resin dry-type transformers comply with VDE 0532-76-11, IEC 60076-11/DIN EN 60076-11 and DIN EN 50541-1. On request, other standards such as GOST, SABS or CSA/ANSI/IEEE, can also be met account.

EU guidelines: Ecodesign Directive of the European Commission

Effective from July 1, 2015, transformers that are installed within the European Economic Area (EEA) must meet the ecodesign requirements of the new directive, provided that they fall within the scope of the directive.
Since the directive is a measure to implement Ecodesign Directive 209/125/EC, the CE mark is used as evidence of compliance. GEAFOL Basic transformers are designed accordingly, and are particularly low-loss and economic.

Characteristic properties (fig. 5.11-5)

HV winding

The high-voltage windings are wound from aluminum foil (copper as winding material on request) interleaved with high-grade insulating foils. The assembled and connected individual coils are placed in a heated mold, and are potted in a vacuum furnace with a mixture of pure silica (quartz sand) and specially blended epoxy resins. The only connections to the outside are casted brass nuts that are internally bonded to the aluminum winding connections.

The external delta connections are made of insulated copper or aluminum connectors to guarantee an optimal installation design. The resulting high-voltage windings are fire-resistant, moisture-proof and corrosion-proof. They show excellent aging properties under all operating conditions.

The foil windings combine a simple winding technique with a high degree of electrical safety. The insulation is subjected to less electrical stress than in other types of windings. In a conventional round-wire winding, the interturn voltages can add up to twice the interlayer voltage. In a foil winding, it never exceeds the voltage per turn, because a layer consists of only one winding turn. This results in high AC voltage and impulse voltage withstand capacity (fig. 5.11-6).

One reason for using aluminum is because the thermal expansion coefficients of aluminum and cast resin are so similar that thermal stresses resulting from load changes are kept to a minimum. However, copper windings are also available.

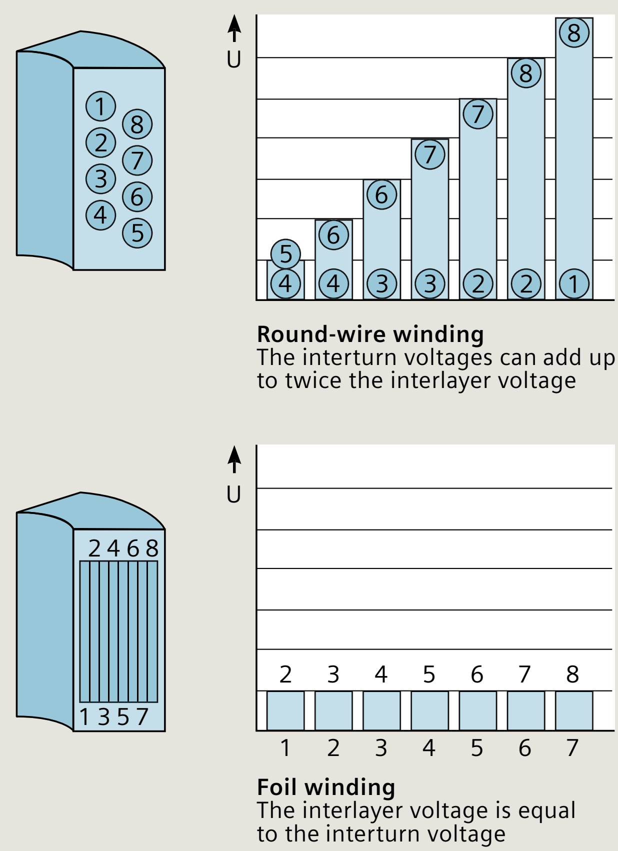

Fig. 5.11-6: High-voltage encapsulated winding design of GEAFOL cast-resin transformer, and voltage stress of a conventional round-wire winding (above) and the foil winding (below)

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LV winding

The standard low-voltage winding with its considerably reduced dielectric stresses is wound from single aluminum sheets (copper as winding material on request) with epoxy-resin pre-impregnated fiberglass fabrics (Prepreg).

The assembled coils are then oven-cured to form uniformly bonded solid cylinders that are impervious to moisture. Through the single-sheet winding design, excellent dynamic stability under short-circuit conditions is achieved. Connections are submerged arc-welded to the aluminum sheets, and are extended either as aluminum or copper bars to the secondary terminals.

Fire safety

GEAFOL transformers use only flame-retardant and self-extinguishing materials in their construction. No additional substances, such as aluminum oxide trihydrate, which could negatively influence the mechanical stability of the cast-resin molding material, are used. Internal arcing from electrical faults, as well as externally applied flames do not cause the transformers to burst or burn. After the source of ignition is removed, the transformer is self-extinguishing. This design has been approved by fire officials in many countries for installation in populated buildings and other structures. The environmental safety of the combustion residues has been proven in many tests (fig. 5.11-7).

Classification of cast-resin transformers

Dry-type transformers have to be classified under the classes listed below:
- Environmental class
- Climatic class
- Fire behavior class.

These classes have to be shown on the rating plate of each dry-type transformer.

The properties laid down in the standards for ratings within the class relating to environment (humidity), climate, and fire behavior have to be demonstrated by means of tests.

These tests are described for the environmental class (code numbers E0, E1 and E2) and for the climatic class (code numbers C1 and C2) in IEC 60076-11. According to this standard, the tests are to be carried out on complete transformers. The tests of fire behavior (fire behavior class code numbers F0 and F1) are limited to tests on a duplicate of a complete transformer that consists of a core leg, a low-voltage winding, and a high-voltage winding.

GEAFOL cast-resin transformers meet the requirements of the highest defined protection classes:
- Environmental class E2 (optionally E3 according to IEC 60076-16, wind turbines application)
- Climatic class C2 (on request, designs for ambient air temperature below -25 °C are available)
- Fire behavior class F1.

Fig. 5.11-7: Flammability test of cast-resin transformer

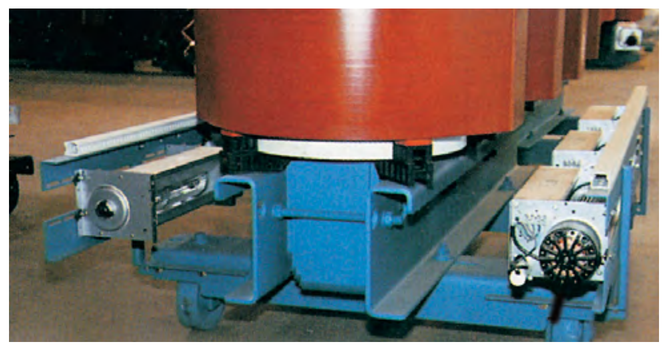

Fig. 5.11-8: Radial cooling fans on GEAFOL transformer for AF cooling

U_m (kV)	*LI* (kV) *2)	*AC* (kV) *2)
1.1	–	3
12	75	28
24	95/125	50
36	145/170	70

*2) *other levels upon request*

Table 5.11-2: Standard insulation levels of GEAFOL (IEC)

Insulation class and temperature rise

The high-voltage winding and the low-voltage winding utilize class F insulating materials with a mean temperature rise of 100 K (standard design).

Overload capability

GEAFOL transformers can be overloaded permanently up to 50 % (with a corresponding increase in impedance voltage and load losses) if additional radial cooling fans are installed (dimensions can increase by approximately 100 mm in length and width) (fig. 5.11-8). Short-time overloads are uncritical as long as the maximum winding temperatures are not exceeded for extended periods of time (depending on initial load and ambient air temperature).

Temperature monitoring
Each GEAFOL transformer is fitted with three temperature sensors installed in the LV winding, and a solid-state tripping device with relay output. The PTC thermistors used for sensing are selected for the applicable maximum hot-spot winding temperature.

Additional sets of sensors can be installed, e.g., for fan control purposes. Alternatively, Pt100 sensors are available. For operating voltages of the LV winding of 3.6 kV and higher, special temperature measuring equipment can be provided.

Auxiliary wiring is run in a protective conduit and terminated in a central LV terminal box (optional). Each wire and terminal is identified, and a wiring diagram is permanently attached to the inside cover of this terminal box.

Installation and enclosures
Indoor installation in electrical operating rooms or in various sheet metal enclosures is the preferred method of installation. The transformers need to be protected against access to the terminals or the winding surfaces, against direct sunlight and against water. Unless sufficient ventilation is provided by the installation location or the enclosure, forced-air cooling must be specified or provided by others.

Instead of the standard open terminals, plug-type elbow connectors can be supplied for the high-voltage side with LI ratings up to 170 kV. Primary cables are usually fed to the transformer from trenches below, but can also be connected from above (fig. 5.11-9).

Secondary connections can be made by multiple insulated cables, or by connecting bars from either below or above. Terminals are made of aluminum (copper upon request).

A variety of indoor and outdoor enclosures in different protection classes are available for the transformers alone, or for indoor compact substations in conjunction with high-voltage and low-voltage switchgear panels. PEHLA-tested housings are also available (fig. 5.11-10).

Using vacuum switches with GEAFOL transformers

Transformers are the key to operating elements at hubs in the distribution system. Vacuum switches control the switching of distribution transformers reliably and safely, with no need for overvoltage protection.

An important parameter in transformers is the magnetization current, one of the “small inductive currents.” Interrupting these currents naturally creates marked transients, but no unacceptably high switching overvoltages, that would pose a threat to connected distribution transformers, are permitted.

Extensive trials using a combination of Siemens GEAFOL transformers and vacuum switches have proven that the

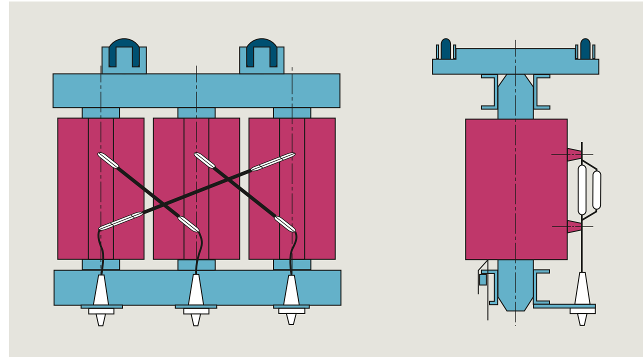

Fig. 5.11-9: GEAFOL transformer with plug-type cable connections

Fig. 5.11-10: GEAFOL transformer in protective housing to IP20/40

GEAFOL medium-voltage windings can handle switching overvoltages with no difficulty – providing further proof of their high product quality and operational safety.

Cost-effective recycling
The oldest of the GEAFOL cast-resin transformers that entered production in the mid-1960s are approaching the end of their service life. Much experience has been gathered over the years with the processing of faulty or damaged coils from such transformers. The metals and resin used in GEAFOL cast-resin transformers, approximately 95 % of their total mass, can be recyled. The process used is non-polluting. Given the value of secondary raw materials, the procedure is often cost-effective, even with the small amounts currently being processed.

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The GEAFOL Basic – a true GEAFOL and more

The GEAFOL Basic is based on almost 50 years of proven GEAFOL technology and quality, but it offers numerous innovations that has allowed Siemens to provide it with several very special characteristics. For example, the GEAFOL Basic distribution transformer and static converter with a maximum rated power of 50 MVA and a maximum medium voltage of 41.4 kV is almost 10 % lighter than a comparable model from the proven GEAFOL series. And this "slimming down" also positively affects the dimensions. This could be achieved by a considerably improved heat dissipation because of the newly patented design. Because of the reduced dimensions, it is predestined for installation in areas with limited space, such as nacelles and towers of wind turbines, data centers, or the upper floors of high-rise buildings.

Of course, all GEAFOL Basic distribution transformers meet the specifications of VDE 0532-76-11/IEC 60076-11/ DIN EN 60076-11 and DIN EN 50541-1. They meet the highest requirements for safe installation in residential and work environments with climatic class C2, environmental class E2 and fire behavior class F1. With fewer horizontal surfaces, less dust is deposited, which leads to a further reduction in the already minimal time and effort needed for maintenance and also increases operational reliability.

Optimum combination

The GEAFOL Basic distribution transformer represents an optimum compromise between performance, safety, and small dimensions. In addition, the high degree of standardization ensures the best possible cost-benefit ratio. Thanks to their compact shape and comprehensive safety certification, GEAFOL Basic distribution transformers can be used in almost every environment.

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A new design for your success –
the reliable, space-saving GEAFOL Basic

1 **Three-limb core** made of grain-oriented, low-loss electric sheet steel insulated on both sides

2 **Low-voltage winding** made of aluminum strip; turns are permanently bonded with insulating sheet

3 **High-voltage winding** made of individual aluminum coils using foil technology and vacuum casting

4 **Low-voltage connectors (facing up)**

5 **Lifting eyes** integrated into the upper core frame for simple transport

6 **Delta connection tubes with HV terminals**

7 **Clamping frame and truck** Convertible rollers for longitudinal and transverse travel

8 **Insulation made of an epoxy resin/quartz powder mixture** makes the transformer extensively maintenance-free, moisture-proof and suitable for the tropics, fire-resistant and self-extinguishing

9 **High-voltage tappings ±2 x 2.5 %** (on the high-voltage terminal side) to adapt to the respective network conditions; reconnectable off load

Temperature monitoring with PTC thermistor detector in limb V of the low-voltage winding (in all three phases on request)

Painting of steel parts
High-build coating, RAL 5009 on request: two-component coating (for particularly aggressive environments)

Structure made of individual components
For example, windings can be individually assembled and replaced on site

Climatic class C2

Environmental class E2

Fire behavior class F1

5.11.4 GEAFOL special transformers

GEAFOL cast-resin transformers with oil-free on-load tap changers (OLTC)

The voltage-regulating cast-resin transformers connected on the load side of the medium-voltage power supply system feed the plant-side distribution transformers. The on-load tap changer controlled transformers used in these medium-voltage systems need to have appropriately high ratings.

Siemens offers suitable transformers with OLTC in its GEAFOL design (fig. 5.11-11), which has proved successful over many years and is available in ratings of up to 50 MVA. The range of rated voltage extends to 36 kV, and the maximum impulse voltage is 200 kV. The main applications of this type of transformer are in modern industrial plants, hospitals, office and apartment blocks, and shopping centers.

Linking 1-pole tap changer modules together by means of insulating shafts produces a 3-pole on-load tap changer for regulating the output voltage of 3-phase GEAFOL transformers. In its nine operating positions, this type of tap changer has a rated current of 500 A and a rated voltage of 900 V per step. This allows voltage fluctuations of up to 7,200 V to be kept under control. However, the maximum control range utilizes only 20 % of the rated voltage.

Transformers for static converters

These are special cast-resin power transformers that are designed for the special demands of thyristor converter or diode rectifier operation.

Fig. 5.11-12: 23 MVA GEAFOL cast-resin transformer 10 kV/Dd0Dy11

The effects of such conversion equipment on transformers and additional construction requirements are as follows:

- Increased load by harmonic currents
- Balancing of phase currents in multiple winding systems (e.g., 12-pulse systems)
- Overload capability
- Higher voltage stress caused by commutation of the thyristors
- Types for 12-pulse systems or higher, if required.

Siemens supplies oil-filled converter transformers of all ratings and configurations known today, and dry-type cast-resin converter transformers up to 50 MVA and 250 kV LI (fig. 5.11-12).

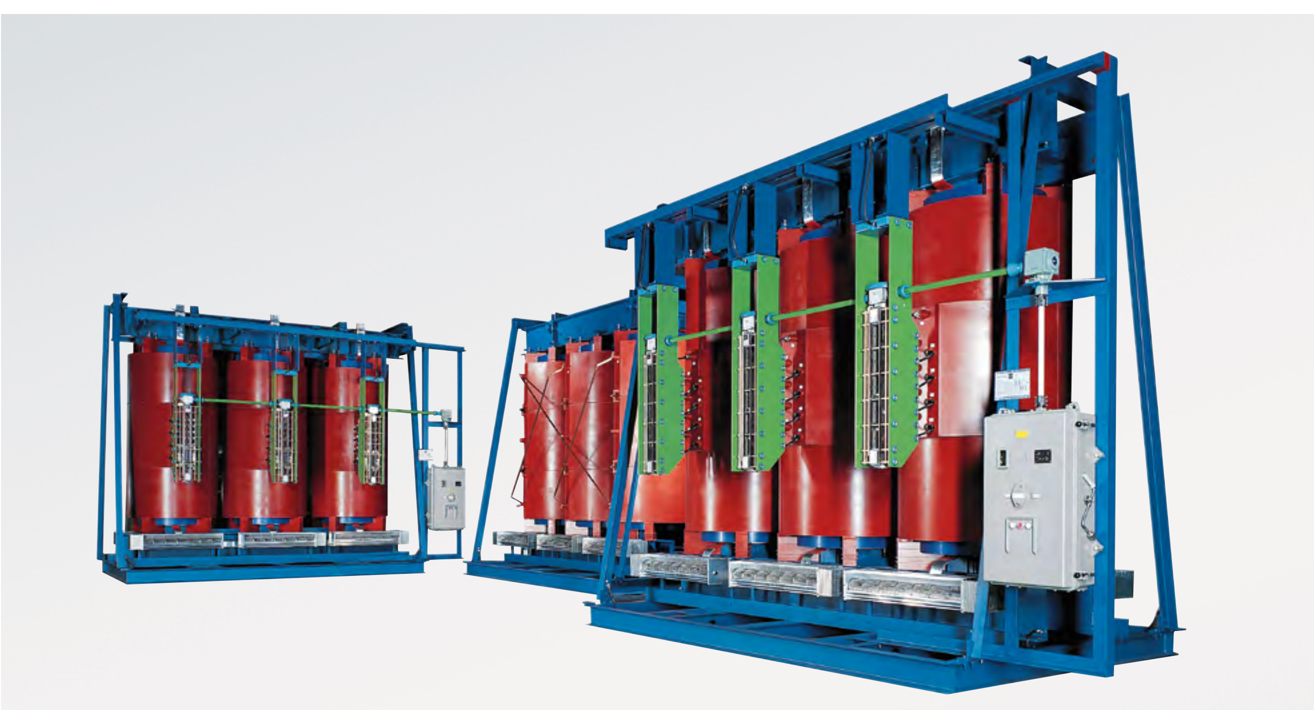

Fig. 5.11-11: 16/22-MVA GEAFOL cast-resin transformer with oil-free on-load tap changer

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To define and quote for such transformers, it is necessary to know considerable details on the converter to be supplied and on the existing harmonics. These transformers are almost exclusively inquired together with the respective drive or rectifier system, and are always custom-engineered for the given application.

Neutral earthing transformers

When a neutral earthing reactor or earth-fault neutralizer is required in a 3-phase system and no suitable neutral is available, a neutral earthing must be provided by using a neutral earthing transformer.

Neutral earthing transformers are available for continuous operation or short-time operation. The zero impedance is normally low. The standard vector group is wye/delta. Some other vector groups are also possible.

Neutral earthing transformers can be built by Siemens in all common power ratings.

Transformers for silicon-reactor power feeding

These special transformers are an important component in plants for producing polycrystalline silicon, which is needed particularly by the solar industry for the manufacture of collectors.

What is special about these transformers is that they have to provide five or more secondary voltages for the voltage supply of the special thyristor controllers. The load is highly unbalanced and is subject to harmonics that are generated by the converters. Special GEAFOL cast-resin transformers with open secondary circuit have been developed for this purpose. The rated power can be up to round about 10 MVA, and the current can exceed an intensity of 5,000 A depending on the reactor type and operating mode. Depending on the reactor control system, two-winding or multi-winding transformers will be used (fig. 5.11-13).

Fig. 5.11-13: 4771 kVA GEAFOL converter transformer with 5 secondary tappings 10/0.33 – 2.4 kV

GEAFOL cast-resin transformers in protective housings with air-water cooling system

GEAFOL cast-resin transformers (fig. 5.11-14) are designed using the special AFWF cooling system. With this system, the thermal losses generated in the windings and in the iron core are not released directly into the environment as hot air, but are collected in a largely airtight protective housing around and above the transformer. Then, they are compressed using fans via an air-water heat exchanger, and released from there into an external cold water circulation system. The re-cooled, cold air is then distributed to all phases using a system of air guide plates, and is fed back to cool the windings from below. A double-pipe construction system for the coolers with leak monitoring ensures additional operational safety. The housing-transformer system is widely used on ships, and is available up to the highest ratings.

Fig. 5.11-14: GEAFOL cast-resin transformers in protective housing with air-water cooling system

GEAFOL cast-resin transformers for wind turbines

Years of electrical, technical and thermal know-how has enabled Siemens to build transformers that are suitable for both onshore and extreme offshore conditions. Installed in the nacelle of the wind turbine, GEAFOL cast-resin transformers require minimal installation effort and offer a range of various network concepts for optimal system efficiency. Various high-voltage and low-voltage connection options allow Siemens GEAFOL transformers to tailor units to meet customers specific needs. They meet the highest fire protection and environmental requirements and ensure maximum operating safety. Additionally, minimum maintenance is required and they are extremely easy to repair.

1 2 3 4 5 6 7 8 9 10 11 12 ?

Rated power S_r kVA	Rated primary voltage[1] tapping ± 2 x 2.5 % U_r HV kV	Rated secondary voltage[3] (no-load) U_r LV kV	Insulation level HV (AC/LI) kV	Insulation level LV (AC/LI) kV	Impedance voltage at rated current u_{zr} %	No-load losses P_o W	Load losses at 120 °C P_{k120} W	Noise level L_{WA} dB	Order No.	Total weight approx. kg	Length A1 mm	Width B1 mm	Height H1 mm
100	10	0.4	28/75	3/–	4	440	1,850	59	4GB5044-3CY05-0AA2	600	1,210	670	840
	10	0.4	28/75	3/–	4	320	1,850	51	4GB5044-3GY05-0AA2	720	1,230	675	845
	10	0.4	28/v75	3/–	6	360	2,000	59	4GB5044-3DY05-0AA2	570	1,200	680	805
	10	0.4	28/75	3/–	6	290	2,000	51	4GB5044-3HY05-0AA2	720	1,280	685	890
	20	0.4	50/95	3/–	4	600	1,750	59	4GB5064-3CY05-0AA2	620	1,220	740	925
	20	0.4	50/95	3/–	4	400	1,750	51	4GB5064-3GY05-0AA2	740	1,260	745	945
	20	0.4	50/95	3/–	6	460	2,050	59	4GB5064-3DY05-0AA2	610	1,250	750	915
	20	0.4	50/95	3/–	6	340	2,050	51	4GB5064-3HY05-0AA2	730	1,280	750	940
	20	0.4	50/125	3/–	6	460	2,050	59	4GB5067-3DY05-0AA2	720	1,260	750	1,145
160	10	0.4	28/75	3/–	4	610	2,600	62	4GB5244-3CY05-0AA2	820	1,270	690	1,025
	10	0.4	28/75	3/–	4	440	2,600	54	4GB5244-3GY05-0AA2	960	1,260	685	1,100
	10	0.4	28/75	3/–	6	500	2,750	62	4GB5244-3DY05-0AA2	690	1,220	685	990
	10	0.4	28/75	3/–	6	400	2,750	54	4GB5244-3HY05-0AA2	850	1,290	695	1,010
	20	0.4	50/95	3/–	4	870	2,500	62	4GB5264-3CY05-0AA2	790	1,280	745	1,060
	20	0.4	50/95	3/–	4	580	2,500	54	4GB5264-3GY05-0AA2	920	1,320	755	1,060
	20	0.4	50/95	3/–	6	650	2,700	62	4GB5264-3DY05-0AA2	780	1,320	760	1,040
	20	0.4	50/95	3/–	6	480	2,700	54	4GB5264-3HY05-0AA2	860	1,350	765	1,050
	20	0.4	50/125	3/–	6	650	2,900	62	4GB5267-3DY05-0AA2	870	1,310	720	1,200
250	10	0.4	28/75	3/–	4	820	3,200	65	4GB5444-3CY05-0AA2	1,010	1,330	700	1,055
	10	0.4	28/75	3/–	4	600	3,200	57	4GB5444-3GY05-0AA2	1,250	1,340	700	1,190
	10	0.4	28/75	3/–	6	700	3,300	65	4GB5444-3DY05-0AA2	960	1,340	705	1,055
	10	0.4	28/75	3/–	6	560	3,300	57	4GB5444-3HY05-0AA2	1,130	1,390	715	1,070
	20	0.4	50/95	3/–	4	1,100	3,200	65	4GB5464-3CY05-0AA2	1,070	1,370	730	1,115
	20	0.4	50/95	3/–	4	800	3,300	57	4GB5464-3GY05-0AA2	1,230	1,420	740	1,130
	20	0.4	50/95	3/–	6	880	3,400	65	4GB5464-3DY05-0AA2	1,020	1,390	740	1,105
	20	0.4	50/95	3/–	6	650	3,400	57	4GB5464-3HY05-0AA2	1,190	1,430	745	1,125
	20	0.4	50/125	3/–	6	880	3,800	65	4GB5467-3DY05-0AA2	1,070	1,390	740	1,200
	30	0.4	70/145	3/–	6	1,280	4,000	67	4GB5475-3DY05-0AA2	1,190	1,450	825	1,365
(315)[4]	10	0.4	28/75	3/–	4	980	3,500	67	4GB5544-3CY05-0AA2	1,120	1,340	820	1,130
	10	0.4	28/75	3/–	4	730	3,500	59	4GB5544-3GY05-0AA2	1,400	1,400	820	1,195
	10	0.4	28/75	3/–	6	850	3,900	67	4GB5544-3DY05-0AA2	1,130	1,360	820	1,160
	10	0.4	28/75	3/–	6	670	3,700	59	4GB5544-3HY05-0AA2	1,260	1,400	820	1,170
	20	0.4	50/95	3/–	4	1,250	3,500	67	4GB5564-3CY05-0AA2	1,370	1,490	835	1,145
	20	0.4	50/95	3/–	4	930	3,500	59	4GB5564-3GY05-0AA2	1,590	1,520	835	1,205
	20	0.4	50/95	3/–	6	1,000	3,800	67	4GB5564-3DY05-0AA2	1,350	1,490	835	1,180
	20	0.4	50/95	3/–	6	780	3,800	59	4GB5564-3HY05-0AA2	1,450	1,520	840	1,205
	20	0.4	50/125	3/–	6	1,000	4,200	67	4GB5567-3DY05-0AA2	1,430	1,520	840	1,235
	30	0.4	70/145	3/–	6	1,450	4,700	69	4GB5575-3DY05-0AA2	1,460	1,510	915	1,445

[1] Applies to U_r HV:
10 to 12 kV
20 to 24 kV
30 to 36 kV

[2] Dimension drawing: fig. 5.11-15, indications are approximate values
[3] Indication of 0.4 kV applies to the voltage range of 0.4–0.5 kV
[4] Ratings in brackets are not standardized

GEAFOL cast-resin transformers comply with IEC 60076-11, DIN EN 60076-11, EN50541-1 and VDE 0532-76-11 without housing, vector group Dyn5, 50 Hz, rated power > 3150 kVA and are not standardized. Other versions and special equipment on request.

Table 5.11-3: GEAFOL cast-resin transformers 100 to 16,000 kVA standard losses (part 1)

Rated power	Rated primary voltage[1] tapping ± 2 x 2.5 %	Rated secondary voltage[3] (no-load)	Insulation level HV (AC/LI)	Insulation level LV (AC/LI)	Impedance voltage at rated current	No-load losses	Load losses at 120 °C	Noise level	Order No.	Total weight	Length	Width	Height
S_r kVA	U_r HV kV	U_r LV kV	kV	kV	u_{zr} %	P_o W	P_{k120} W	L_{WA} dB		approx. kg	A1 mm	B1 mm	H1 mm
400	10	0.4	28/75	3/–	4	1,150	4,400	68	4GB5644-3CY05-0AA2	1,290	1,370	820	1,230
	10	0.4	28/75	3/–	4	880	4,400	60	4GB5644-3GY05-0AA2	1,500	1,390	820	1,330
	10	0.4	28/75	3/–	6	1,000	4,900	68	4GB5644-3DY05-0AA2	1,230	1,400	820	1,215
	10	0.4	28/75	3/–	6	800	4,900	60	4GB5644-3HY05-0AA2	1,390	1,430	820	1,230
	20	0.4	50/95	3/–	4	1,450	3,800	68	4GB5664-3CY05-0AA2	1,470	1,460	830	1,285
	20	0.4	50/95	3/–	4	1,100	3,800	60	4GB5664-3GY05-0AA2	1,710	1,520	835	1,305
	20	0.4	50/95	3/–	6	1,200	4,300	68	4GB5664-3DY05-0AA2	1,380	1,490	835	1,260
	20	0.4	50/95	3/–	6	940	4,300	60	4GB5664-3HY05-0AA2	1,460	1,500	840	1,260
	20	0.4	50/125	3/–	6	1,200	4,700	68	4GB5667-3DY05-0AA2	1,530	1,540	845	1,310
	30	0.4	70/145	3/–	6	1,650	5,500	69	4GB5675-3DY05-0AA2	1,590	1,560	925	1,500
(500)[4]	10	0.4	28/75	3/–	4	1,300	5,900	69	4GB5744-3CY05-0AA0	1,490	1,410	820	1,315
	10	0.4	28/75	3/–	4	1,000	5,300	61	4GB5744-3GY05-0AA0	1,620	1,420	820	1,340
	10	0.4	28/75	3/–	6	1,200	6,400	69	4GB5744-3DY05-0AA0	1,420	1,450	820	1,245
	10	0.4	28/75	3/–	6	950	6,400	61	4GB5744-3HY05-0AA0	1,540	1,490	820	1,265
	20	0.4	50/95	3/–	4	1,700	4,900	69	4GB5764-3CY05-0AA0	1,550	1,460	840	1,365
	20	0.4	50/95	3/–	4	1,300	4,900	61	4GB5764-3GY05-0AA0	1,700	1,490	845	1,370
	20	0.4	50/95	3/–	6	1,400	5,100	69	4GB5764-3DY05-0AA0	1,500	1,530	855	1,275
	20	0.4	50/95	3/–	6	1,100	5,100	61	4GB5764-3HY05-0AA0	1,670	1,560	860	1,290
	20	0.4	50/125	3/–	6	1,400	6,300	69	4GB5767-3DY05-0AA0	1,610	1,540	855	1,355
	30	0.4	70/145	3/–	6	1,900	6,000	70	4GB5775-3DY05-0AA0	1,810	1,560	925	1,615
	30	0.4	70/170	3/–	6	2,600	6,200	79	4GB5780-3DY05-0AA0	2,110	1,710	1,005	1,590
630	10	0.4	28/75	3/–	4	1,500	7,300	70	4GB5844-3CY05-0AA0	1,670	1,410	820	1,485
	10	0.4	28/75	3/–	4	1,150	7,300	62	4GB5844-3GY05-0AA0	1,840	1,440	820	1,485
	10	0.4	28/75	3/–	6	1,370	7,500	70	4GB5844-3DY05-0AA0	1,710	1,520	830	1,305
	10	0.4	28/75	3/–	6	1,100	7,500	62	4GB5844-3HY05-0AA0	1,850	1,560	835	1,330
	20	0.4	50/95	3/–	4	2,000	6,900	70	4GB5864-3CY05-0AA0	1,790	1,470	840	1,530
	20	0.4	50/95	3/–	4	1,600	6,900	62	4GB5864-3GY05-0AA0	1,930	1,520	845	1,565
	20	0.4	50/95	3/–	6	1,650	6,800	70	4GB5864-3DY05-0AA0	1,750	1,560	860	1,365
	20	0.4	50/95	3/–	6	1,250	6,800	62	4GB5864-3HY05-0AA0	1,900	1,600	865	1,385
	20	0.4	50/125	3/–	6	1,650	7,000	70	4GB5867-3DY05-0AA0	1,830	1,590	865	1,395
	30	0.4	70/145	3/–	6	2,200	6,600	71	4GB5875-3DY05-0AA0	2,090	1,620	940	1,640
800	10	0.4	28/75	3/–	4	1,800	7,800	72	4GB5944-3CY05-0AA0	1,970	1,500	820	1,535
	10	0.4	28/75	3/–	4	1,400	7,800	64	4GB5944-3GY05-0AA0	2,210	1,530	825	1,535
	10	0.4	28/75	3/–	6	1,700	8,300	72	4GB5944-3DY05-0AA0	2,020	1,590	840	1,395
	10	0.4	28/75	3/–	6	1,300	8,300	64	4GB5944-3HY05-0AA0	2,230	1,620	845	1,395
	20	0.4	50/95	3/–	4	2,400	8,500	72	4GB5964-3CY05-0AA0	2,020	1,550	850	1,595
	20	0.4	50/95	3/–	4	1,900	8,500	64	4GB5964-3GY05-0AA0	2,220	1,570	855	1,595
	20	0.4	50/95	3/–	6	1,900	8,200	72	4GB5964-3DY05-0AA0	2,020	1,610	870	1,435
	20	0.4	50/95	3/–	6	1,500	8,200	64	4GB5964-3HY05-0AA0	2,220	1,650	875	1,455
	20	0.4	50/125	3/–	6	1,900	9,400	72	4GB5967-3DY05-0AA0	2,160	1,660	880	1,485
	30	0.4	70/145	3/–	6	2,650	7,900	72	4GB5975-3DY05-0AA0	2,620	1,740	965	1,695

1) Applies to Ur HV:
10 to 12 kV
20 to 24 kV
30 to 36 kV

2) Dimension drawing: fig. 5.11-15, indications are approximate values
3) Indication of 0.4 kV applies to the voltage range of 0.4–0.5 kV
4) Ratings in brackets are not standardized

GEAFOL cast-resin transformers comply with IEC 60076-11, DIN EN 60076-11, EN50541-1 and VDE 0532-76-11 without housing, vector group Dyn5, 50 Hz, rated power > 3150 kVA and are not standardized. Other versions and special equipment on request.

Table 5.11-3: GEAFOL cast-resin transformers 100 to 16,000 kVA standard losses (part 2)

Rated power	Rated primary voltage[1] tapping ± 2 x 2.5 %	Rated secondary voltage[3] (no-load)	Insulation level HV (AC/LI)	Insulation level LV (AC/LI)	Impedance voltage at rated current	No-load losses	Load losses at 120 °C	Noise level	Order No.	Total weight	Length	Width	Height
S_r kVA	U_r HV kV	U_r LV kV	kV	kV	u_{zr} %	P_o W	P_{k120} W	L_{WA} dB		approx. kg	A1 mm	B1 mm	H1 mm
1,000	10	0.4	28/75	3/–	4	2,100	10,000	73	4GB6044-3CY05-0AA0	2,440	1,550	990	1,730
	10	0.4	28/75	3/–	4	1,600	10,000	65	4GB6044-3GY05-0AA0	2,850	1,620	990	1,795
	10	0.4	28/75	3/–	6	2,000	9,500	73	4GB6044-3DY05-0AA0	2,370	1,640	990	1,490
	10	0.4	28/75	3/–	6	1,500	9,500	65	4GB6044-3HY05-0AA0	2,840	1,710	990	1,565
	20	0.4	50/95	3/–	4	2,800	9,500	73	4GB6064-3CY05-0AA0	2,420	1,570	990	1,790
	20	0.4	50/95	3/–	4	2,300	8,700	65	4GB6064-3GY05-0AA0	2,740	1,680	990	1,665
	20	0.4	50/95	3/–	6	2,300	9,400	73	4GB6064-3DY05-0AA0	2,310	1,640	990	1,620
	20	0.4	50/95	3/–	6	1,800	9,400	65	4GB6064-3HY05-0AA0	2,510	1,660	990	1,620
	20	0.4	50/125	3/–	6	2,300	11,000	73	4GB6067-3DY05-0AA0	2,470	1,670	990	1,650
	30	0.4	70/145	3/–	6	3,100	10,000	73	4GB6075-3DY05-0AA0	2,990	1,800	1,060	1,795
(1,250)[4]	10	0.4	28/75	3/–	6	2,400	11,000	75	4GB6144-3DY05-0AA0	2,780	1,740	990	1,635
	10	0.4	28/75	3/–	6	1,800	11,000	67	4GB6144-3HY05-0AA0	3,140	1,770	990	1,675
	20	0.4	50/95	3/–	6	2,700	11,200	75	4GB6164-3DY05-0AA0	2,740	1,780	990	1,645
	20	0.4	50/95	3/–	6	2,100	11,200	67	4GB6164-3HY05-0AA0	3,010	1,810	990	1,645
	20	0.4	50/125	3/–	6	2,700	10,500	75	4GB6167-3DY05-0AA0	2,980	1,810	990	1,675
	30	0.4	70/145	3/–	6	3,600	11,500	75	4GB6175-3DY05-0AA0	3,580	1,870	1,065	1,895
1,600	10	0.4	28/75	3/–	6	2,800	14,000	76	4GB6244-3DY05-0AA0	3,490	1,830	990	1,735
	10	0.4	28/75	3/–	6	2,100	14,000	68	4GB6244-3HY05-0AA0	4,130	1,880	990	1,775
	20	0.4	50/95	3/–	6	3,100	13,500	76	4GB6264-3DY05-0AA0	3,440	1,840	995	1,830
	20	0.4	50/95	3/–	6	2,400	13,500	68	4GB6264-3HY05-0AA0	3,830	1,870	1,000	1,880
	20	0.4	50/125	3/–	6	3,100	12,500	76	4GB6267-3DY05-0AA0	3,690	1,860	995	1,880
	30	0.4	70/145	3/–	6	4,100	13,500	76	4GB6275-3DY05-0AA0	4,350	1,970	1,090	1,995
(2,000)[4]	10	0.4	28/75	3/–	6	3,500	15,700	78	4GB6344-3DY05-0AA0	4,150	1,940	1,280	1,935
	10	0.4	28/75	3/–	6	2,600	15,700	70	4GB6344-3HY05-0AA0	4,890	1,970	1,280	2,015
	20	0.4	50/95	3/–	6	4,000	15,400	78	4GB6364-3DY05-0AA0	4,170	1,980	1,280	1,960
	20	0.4	50/95	3/–	6	2,900	15,400	70	4GB6364-3HY05-0AA0	4,720	2,010	1,280	1,985
	20	0.4	50/125	3/–	6	4,000	15,500	78	4GB6367-3DY05-0AA0	4,430	2,020	1,280	2,005
	30	0.4	70/145	3/–	6	5,000	15,000	78	4GB6375-3DY05-0AG0	5,090	2,100	1,280	2,135
2,500	10	0.4	28/75	3/–	6	4,300	18,700	81	4GB6444-3DY05-0AG0	4,840	2,090	1,280	2,070
	10	0.4	28/75	3/–	6	3,000	18,700	71	4GB6444-3HY05-0AA0	5,940	2,160	1,280	2,135
	20	0.4	50/95	3/–	6	5,000	18,000	81	4GB6464-3DY05-0AA0	5,200	2,150	1,280	2,165
	20	0.4	50/95	3/–	6	3,600	19,000	71	4GB6464-3HY05-0AA0	6,020	2,190	1,280	2,180
	20	0.4	50/125	3/–	6	5,000	18,000	81	4GB6467-3DY05-0AG0	5,020	2,160	1,280	2,105
	30	0.4	70/145	3/–	6	5,800	20,000	81	4GB6475-3DY05-0AG0	5,920	2,280	1,280	2,215

[1] Applies to Ur HV:
10 to 12 kV
20 to 24 kV
30 to 36 kV

[2] Dimension drawing: fig. 5.11-15, indications are approximate values
[3] Indication of 0.4 kV applies to the voltage range of 0.4–0.55 kV
[4] Ratings in brackets are not standardized

GEAFOL cast-resin transformers comply with IEC 60076-11, DIN EN 60076-11, EN50541-1 and VDE 0532-76-11 without housing, vector group Dyn5, 50 Hz, rated power > 3,150 kVA and are not standardized. Other versions and special equipment on request.

Table 5.11-3: GEAFOL cast-resin transformers 100 to 16,000 kVA standard losses (part 3)

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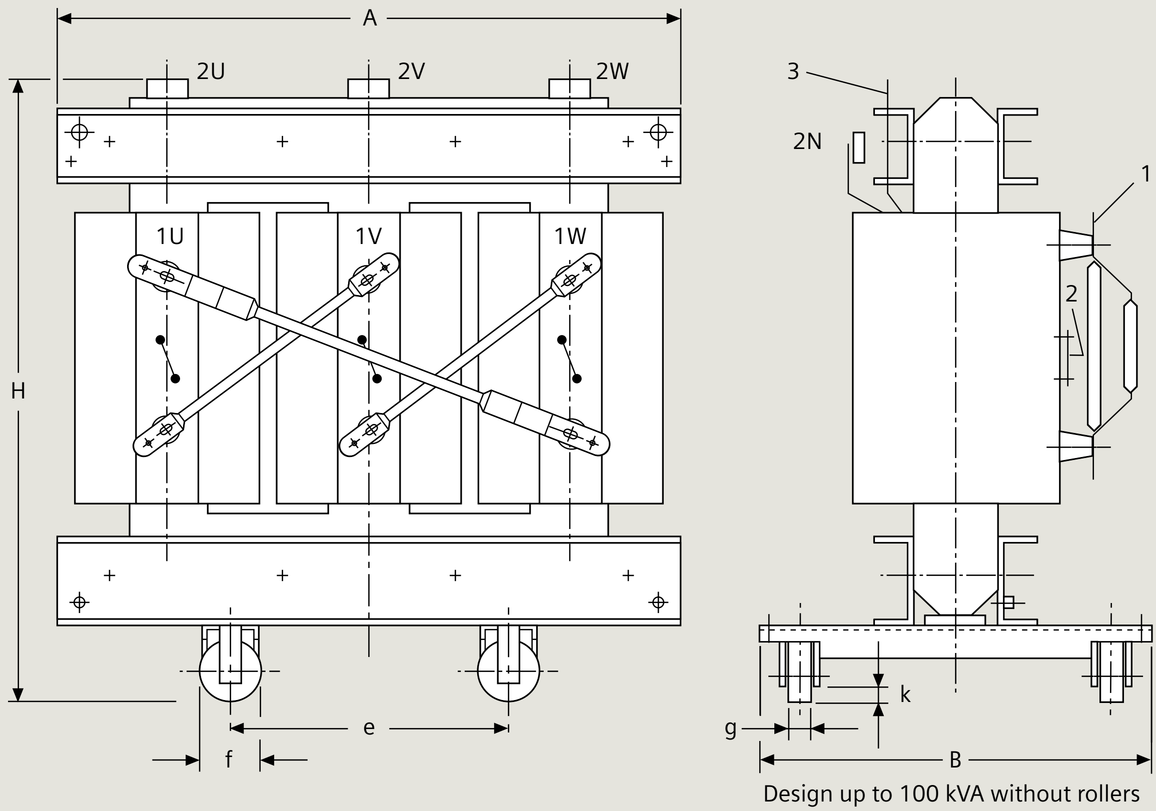


Dimension drawing

Dimensions A, B and H, see tab. 5.11-3,
Dimension e applies to lengthways and sideways travel

1 High-voltage terminals
2 High-voltage tappings on HV side
3 Low-voltage terminals

Fig. 5.11-15: Dimension drawing of GEAFOL cast-resin transformers

Rated power S_r kVA	Rated primary voltage[1)] tapping ± 2 x 2.5 % U_r HV kV	Rated secondary voltage[3)] (no-load) U_r LV kV	Insulation level HV (AC/LI) kV	Insulation level LV (AC/LI) kV	Impedance voltage at rated current u_{zr} %	No-load losses P_o W	Load losses at 120 °C P_{k120} W	Noise level L_{WA} dB	Order No.	Total weight approx. kg	Length[1)] A1 mm	Width[1)] B1 mm	Height[1)] H1 mm
100	10	0.4	28/75	3/–	4	280	2,050	51	4GT50443FY	740	1,220	690	970
	20	0.4	50/95	3/–	4	280	2,050	51	4GT50643FY	860	1,240	750	1,200
160	10	0.4	28/75	3/–	4	400	2,900	54	4GT52443FY	840	1,240	695	1,100
	20	0.4	50/95	3/–	4	400	2,900	54	4GT52643FY	970	1,280	725	1,180
250	10	0.4	28/75	3/–	4	520	3,800	57	4GT54443FY	1,170	1,340	715	1,125
	20	0.4	50/95	3/–	4	520	3,800	57	4GT54643FY	1,380	1,420	750	1,225
315	10	0.4	28/75	3/–	4	650	4,500	59	4GT55443FY	1,280	1,390	820	1,115
	20	0.4	50/95	3/–	4	650	4,500	59	4GT55643FY	1,540	1,490	840	1,215
400	10	0.4	28/75	3/–	4	750	5,500	60	4GT56443FY	1,360	1,370	820	1,270
	20	0.4	50/95	3/–	4	750	5,500	60	4GT56643FY	1,600	1,450	835	1,355
500	10	0.4	28/75	3/–	4	950	6,600	61	4GT57443FY	1,540	1,430	820	1,270
	20	0.4	50/95	3/–	4	950	6,600	61	4GT57643FY	1,770	1,520	845	1,390
630	10	0.4	28/75	3/–	6	1,100	7,600	62	4GT58443EY	1,820	1,500	840	1,485
	20	0.4	50/95	3/–	6	1,100	7,600	62	4GT58643EY	1,860	1,540	870	1,505
800	10	0.4	28/75	3/–	6	1,300	8,000	64	4GT59443EY	2,190	1,600	860	1,505
	20	0.4	50/95	3/–	6	1,300	8,000	64	4GT59643EY	2,520	1,680	900	1,595
1,000	10	0.4	28/75	3/–	6	1,550	9,000	65	4GT60443EY	2,520	1,640	990	1,575
	20	0.4	50/95	3/–	6	1,550	9,000	65	4GT60643EY	2,580	1,690	990	1,635
1,250	10	0.4	28/75	3/–	6	1,800	11,000	67	4GT61443EY	3,030	1,740	990	1,695
	20	0.4	50/95	3/–	6	1,800	11,000	67	4GT61643EY	2,850	1,760	995	1,735
1,600	10	0.4	28/75	3/–	6	2,200	13,000	68	4GT6244 3EY	3,520	1,695	990	1,845
	20	0.4	50/95	3/–	6	2,200	13,000	68	4GT6264 3EY	3,710	1,755	1,005	1,895
2,000	10	0.4	28/75	3/–	6	2,600	16,000	70	4GT63443EY	4,270	1,870	1,280	1,885
	20	0.4	50/95	3/–	6	2,600	16,000	70	4GT63643EY	4,650	1,930	1,280	1,975
2,500	10	0.4	28/75	3/–	6	3,100	19,000	71	4GT64443EY	5,430	2,000	1,280	2,125
	20	0.4	50/95	3/–	6	3,100	19,000	71	4GT64643EY	5,750	2,045	1,280	2,175
3,150	10	0.4	28/75	3/–	6	3,800	22,000	74	4GT65443EY	7,080	2,140	1,280	2,435
	20	0.4	50/95	3/–	6	3,800	22,000	74	4GT65643EY	7,420	2,185	1,280	2,490

[1)] Dimension drawing: fig. 5.11-16, indications are approximate values

All GEAFOL Basic transformers comply with DIN VDE 0532-76-11/DIN EN 60076-11/IEC 60076-11/DIN EN 50541-1. Power ratings >3150 kVA, different voltages and designs, as well as special equipment on request.

Table 5.11-4: GEAFOL Basic transformers according to Ecodesign Directive 2009/125/EG, EU Regulation No. 548/2014

Rated power S_r kVA	Rated primary voltage[1] tapping ± 2 x 2.5 % U_r HV kV	Rated secondary voltage[3] (no-load) U_r LV kV	Insulation level HV (AC/LI) kV	Insulation level LV (AC/LI) kV	Impedance voltage at rated current u_{zr} %	No-load losses P_o W	Load losses at 120 °C P_{k120} W	Noise level L_{WA} dB	Order No.	Total weight approx. kg	Length[1] A1 mm	Width[1] B1 mm	Height[1] H1 mm
100	10	0.4	28/75	3/–	4	440	1,850	61	4GT50443CY	600	1,190	685	920
	10	0.4	28/75	3/–	4	320	1,850	51	4GT50443GY	780	1,230	690	985
	10	0.4	28/75	3/–	6	360	2,000	61	4GT50443DY	580	1,200	690	910
	10	0.4	28/75	3/–	6	290	2,000	51	4GT50443HY	710	1,210	690	1,040
	20	0.4	50/95	3/–	4	600	1,750	61	4GT50643CY	690	1,230	750	1,035
	20	0.4	50/95	3/–	4	400	1,750	51	4GT50643GY	880	1,290	760	1,085
	20	0.4	50/95	3/–	6	460	2,050	61	4GT50643DY	700	1,270	765	1,040
	20	0.4	50/95	3/–	6	340	2,050	51	4GT50643HY	780	1,260	725	1,120
160	10	0.4	28/75	3/–	4	610	2,600	65	4GT52443CY	840	1,270	700	1,005
	10	0.4	28/75	3/–	4	440	2,600	54	4GT52443GY	940	1,270	700	1,105
	10	0.4	28/75	3/–	6	500	2,750	65	4GT52443DY	790	1,280	710	980
	10	0.4	28/75	3/–	6	400	2,750	54	4GT52443HY	860	1,310	710	990
	20	0.4	50/95	3/–	4	700	2,500	65	4GT52643CY	910	1,330	770	1,085
	20	0.4	50/95	3/–	4	580	2,500	54	4GT52643GY	1,070	1,340	735	1,130
	20	0.4	50/95	3/–	6	650	2,700	65	4GT52643DY	850	1,330	775	1,075
	20	0.4	50/95	3/–	6	480	2,700	54	4GT52643HY	950	1,360	745	1,095
250	10	0.4	28/75	3/–	4	820	3,200	68	4GT54443CY	990	1,320	705	1,045
	10	0.4	28/75	3/–	4	600	3,200	57	4GT54443GY	1,140	1,340	710	1,125
	10	0.4	28/75	3/–	6	700	3,300	68	4GT54443DY	940	1,350	715	1,045
	10	0.4	28/75	3/–	6	560	3,300	57	4GT54443HY	1,100	1,380	725	1,070
	20	0.4	50/95	3/–	4	880	3,200	68	4GT54643CY	1,100	1,360	740	1,155
	20	0.4	50/95	3/–	4	800	3,300	57	4GT54643GY	1,290	1,400	745	1,210
	20	0.4	50/95	3/–	6	880	3,400	68	4GT54643DY	1,040	1,400	750	1,115
	20	0.4	50/95	3/–	6	650	3,400	57	4GT54643HY	1,180	1,430	755	1,135
315	10	0.4	28/75	3/–	4	980	3,500	68	4GT55443CY	1,150	1,370	820	1,075
	10	0.4	28/75	3/–	4	730	3,500	59	4GT55443GY	1,350	1,390	820	1,155
	10	0.4	28/75	3/–	6	850	3,900	68	4GT55443DY	1,080	1,370	820	1,120
	10	0.4	28/75	3/–	6	670	3,700	59	4GT55443HY	1,190	1,410	820	1,125
	20	0.4	50/95	3/–	4	1,250	3,500	68	4GT55643CY	1,310	1,440	835	1,190
	20	0.4	50/95	3/–	4	930	3,500	59	4GT55643GY	1,470	1,470	840	1,210
	20	0.4	50/95	3/–	6	1,000	3,800	68	4GT55643DY	1,250	1,450	840	1,200
	20	0.4	50/95	3/–	6	780	3,800	59	4GT55643HY	1,350	1,480	840	1,190
400	10	0.4	28/75	3/–	4	1,150	4,400	68	4GT56443CY	1,320	1,400	820	1,225
	10	0.4	28/75	3/–	4	880	4,400	60	4GT56443GY	1,470	1,390	820	1,325
	10	0.4	28/75	3/–	6	1,000	4,900	68	4GT56443DY	1,250	1,410	820	1,220
	10	0.4	28/75	3/–	6	800	4,900	60	4GT56443HY	1,450	1,460	820	1,240
	20	0.4	50/95	3/–	4	1,270	3,800	68	4GT56643CY	1,460	1,460	840	1,310
	20	0.4	50/95	3/–	4	1,100	3,800	60	4GT56643GY	1,670	1,520	845	1,310
	20	0.4	50/95	3/–	6	1,200	4,300	68	4GT56643DY	1,370	1,480	845	1,275
	20	0.4	50/95	3/–	6	940	4,300	60	4GT56643HY	1,540	1,530	850	1,320
500	10	0.4	28/75	3/–	4	1,300	5,900	69	4GT57443CY	1,480	1,450	820	1,180
	10	0.4	28/75	3/–	4	1,000	5,300	61	4GT57443GY	1,630	1,420	820	1,345
	10	0.4	28/75	3/–	6	1,200	6,400	69	4GT57443DY	1,390	1,450	820	1,255
	10	0.4	28/75	3/–	6	950	6,400	61	4GT57443HY	1,540	1,490	820	1,250
	20	0.4	50/95	3/–	4	1,700	4,900	69	4GT57643CY	1,650	1,510	845	1,370
	20	0.4	50/95	3/–	4	1,300	4,900	61	4GT57643GY	1,830	1,520	845	1,385
	20	0.4	50/95	3/–	6	1,400	5,100	69	4GT57643DY	1,530	1,520	855	1,270
	20	0.4	50/95	3/–	6	1,100	5,100	61	4GT57643HY	1,750	1560	860	1,355

[1] Dimension drawing: fig. 5.11-16, indications are approximate values

All GEAFOL Basic transformers comply with DIN VDE 0532-76-11/DIN EN 60076-11/IEC 60076-11/DIN EN 50541-1. Power ratings >3150 kVA, different voltages and designs, as well as special equipment on request.

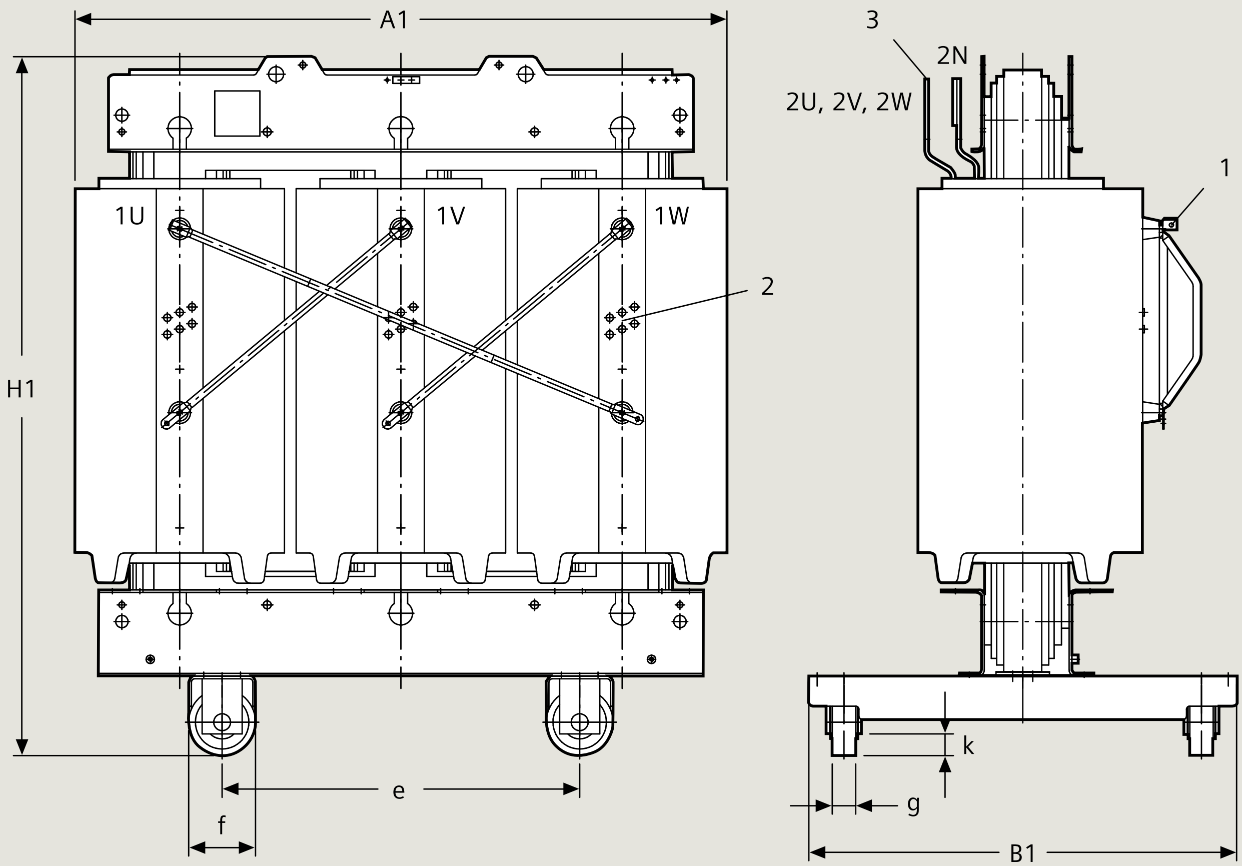
Table 5.11-5: GEAFOL Basic transformers (part 1)

Rated power S_r kVA	Rated primary voltage[1] tapping ± 2 x 2.5 % U_r HV kV	Rated secondary voltage[3] (no-load) U_r LV kV	Insulation level HV (AC/LI) kV	Insulation level LV (AC/LI) kV	Impedance voltage at rated current u_{zr} %	No-load losses P_o W	Load losses at 120 °C P_{k120} W	Noise level L_{WA} dB	Order No.	Total weight approx. kg	Length[1] A1 mm	Width[1] B1 mm	Height[1] H1 mm
630	10	0.4	28/75	3/–	4	1,500	7,300	70	4GT58443CY	1,660	1,460	820	1,335
	10	0.4	28/75	3/–	4	1,150	7,300	62	4GT58443GY	1,880	1,480	820	1,430
	10	0.4	28/75	3/–	6	1,370	7,500	70	4GT58443DY	1,760	1,550	835	1,350
	10	0.4	28/75	3/–	6	1,100	7,500	62	4GT58443HY	1,900	1,560	835	1,370
	20	0.4	50/95	3/–	4	2,000	6,900	70	4GT58643CY	1,900	1,550	855	1,405
	20	0.4	50/95	3/–	4	1,600	6,900	62	4GT58643GY	2,090	1,520	845	1,570
	20	0.4	50/95	3/–	6	1,650	6,800	70	4GT58643DY	1,800	1,590	865	1,370
	20	0.4	50/95	3/–	6	1,250	6,800	62	4GT58643HY	21,40	1,650	880	1,400
800	10	0.4	28/75	3/–	4	1,800	7,800	72	4GT59443CY	2,020	1,520	820	1,520
	10	0.4	28/75	3/–	4	1,400	7,800	64	4GT59443GY	2,280	1,550	830	1,540
	10	0.4	28/75	3/–	6	1,700	8,300	72	4GT59443DY	2,000	1,610	845	1,345
	10	0.4	28/75	3/–	6	1,300	8,300	64	4GT59443HY	2,240	1,640	855	1,370
	20	0.4	50/95	3/–	4	2,400	8,500	72	4GT59643CY	2,140	1,550	855	1,615
	20	0.4	50/95	3/–	4	1,900	8,500	64	4GT59643GY	2,360	1,580	860	1,605
	20	0.4	50/95	3/–	6	1,900	8,200	72	4GT59643DY	2,120	1,650	875	1,445
	20	0.4	50/95	3/–	6	1,500	8,200	64	4GT59643HY	2,340	1,670	885	1,450
1000	10	0.4	28/75	3/–	4	2,100	10,000	73	4GT60443CY	2,560	1,600	990	1,635
	10	0.4	28/75	3/–	4	1,600	10,000	65	4GT60443GY	2,980	1,640	990	1,745
	10	0.4	28/75	3/–	6	2,000	9,500	73	4GT60443DY	2,390	1,620	990	1,580
	10	0.4	28/75	3/–	6	1,500	9,500	65	4GT60443HY	2,650	1,660	990	1,560
	20	0.4	50/95	3/–	4	2,800	9,500	73	4GT60643CY	2,580	1,590	990	1,790
	20	0.4	50/95	3/–	4	2,300	8,700	65	4GT60643GY	2,860	1,630	990	1,810
	20	0.4	50/95	3/–	6	2,300	9,000	73	4GT60643DY	2,460	1,660	990	1,645
	20	0.4	50/95	3/–	6	1,800	9,000	65	4GT60643HY	2,760	1,700	990	1,680
1250	10	0.4	28/75	3/–	6	2,400	11,000	78	4GT61443DY	2,710	1,720	990	1,655
	10	0.4	28/75	3/–	6	1,800	11,000	68	4GT61443HY	3,220	1,780	990	1,715
	20	0.4	50/95	3/–	6	2,700	11,200	78	4GT61643DY	2,850	1,780	990	1,695
	20	0.4	50/95	3/–	6	2,100	11,200	68	4GT61643HY	3,150	1,800	990	1,710
1600	10	0.4	28/75	3/–	6	2,800	13,600	76	4GT62443DY	2,970	1,705	990	1,710
	10	0.4	28/75	3/–	6	2,100	13,600	68	4GT62443HY	3,340	1,745	990	1,730
	20	0.4	50/95	3/–	6	3,100	13,200	76	4GT62643DY	3,200	1,765	1,010	1,810
	20	0.4	50/95	3/–	6	2,400	13,200	68	4GT62643HY	3,570	1,800	1,015	1,860
2000	10	0.4	28/75	3/–	6	3,500	15,500	78	4GT63443DY	3,640	1,805	1,280	1,815
	10	0.4	28/75	3/–	6	2,600	15,500	70	4GT63443HY	4,090	1,855	1,280	1,850
	20	0.4	50/95	3/–	6	3,900	15,800	78	4GT63643DY	3,700	1,785	1,280	2,025
	20	0.4	50/95	3/–	6	2,900	15,800	70	4GT63643HY	4,070	1,820	1,280	2,055
2500	10	0.4	28/75	3/–	6	4,300	20,000	81	4GT64443DY	4,380	1,895	1,280	2,045
	10	0.4	28/75	3/–	6	3,000	20,000	71	4GT64443HY	5,030	1,920	1,280	2,085
	20	0.4	50/95	3/–	6	4,400	19,000	81	4GT64643DY	4,590	1,900	1,280	2,150
	20	0.4	50/95	3/–	6	3,500	19,000	71	4GT64643HY	5,070	1,990	1,280	2,135

[1] Dimension drawing: fig. 5.11-16, indications are approximate values

All GEAFOL Basic transformers comply with DIN VDE 0532-76-11/DIN EN 60076-11/IEC 60076-11/DIN EN 50541-1. Power ratings >3150 kVA, different voltages and designs, as well as special equipment on request.

Table 5.11-5: GEAFOL Basic transformers (part 2)



Dimension drawing

Dimensions A1, B1 and H1, see tab. 5.11-4, 5.11-5
Dimension e applies to longitudinal and transverse travel

1 High-voltage terminals
2 High-voltage tappings on HV terminal side
3 Low-voltage terminals

Fig. 5.11-16: Dimension drawing of GEAFOL Basic transformer

Notes

The technical data, dimensions and weights are subject to change unless otherwise stated on the individual pages of this catalog. The illustrations are for reference only.
All product designations used are trademarks or product names of Siemens AG or of other suppliers.
All dimensions in this catalog are given in mm.
The information in this document contains general descriptions of the technical options available, which do not always have to be present in individual cases. The required features should therefore be specified in each individual case at the time of closing the contract.